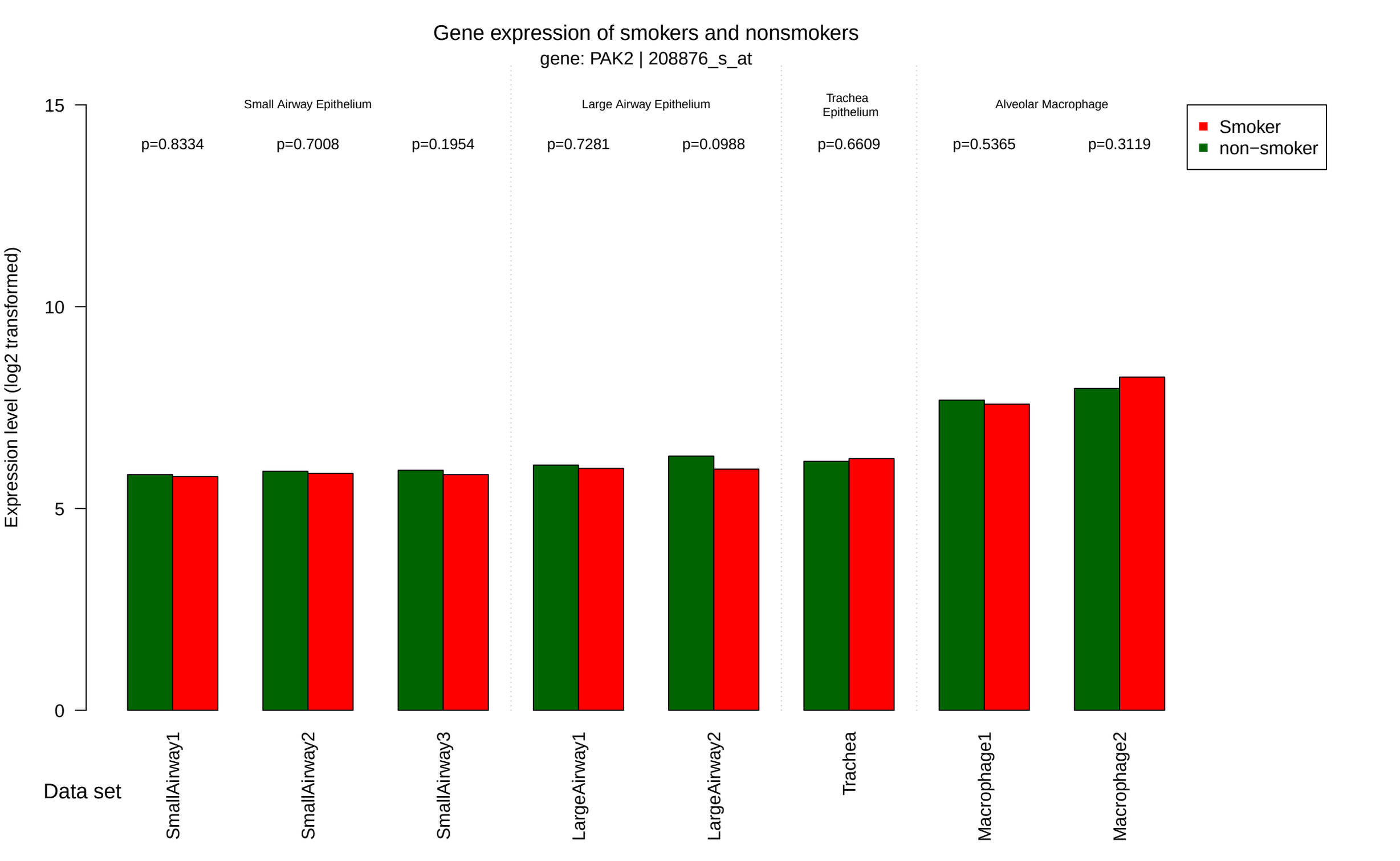
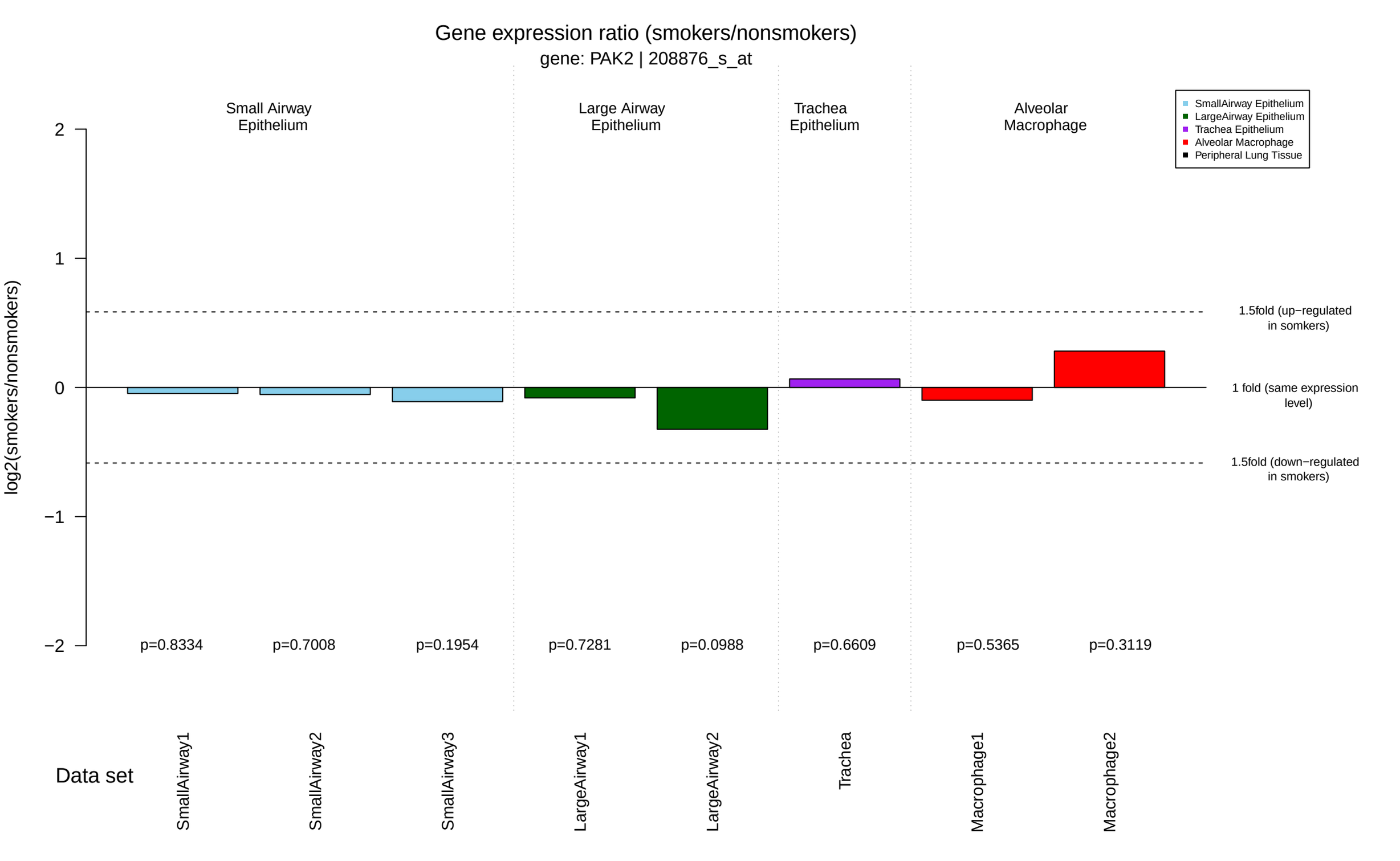
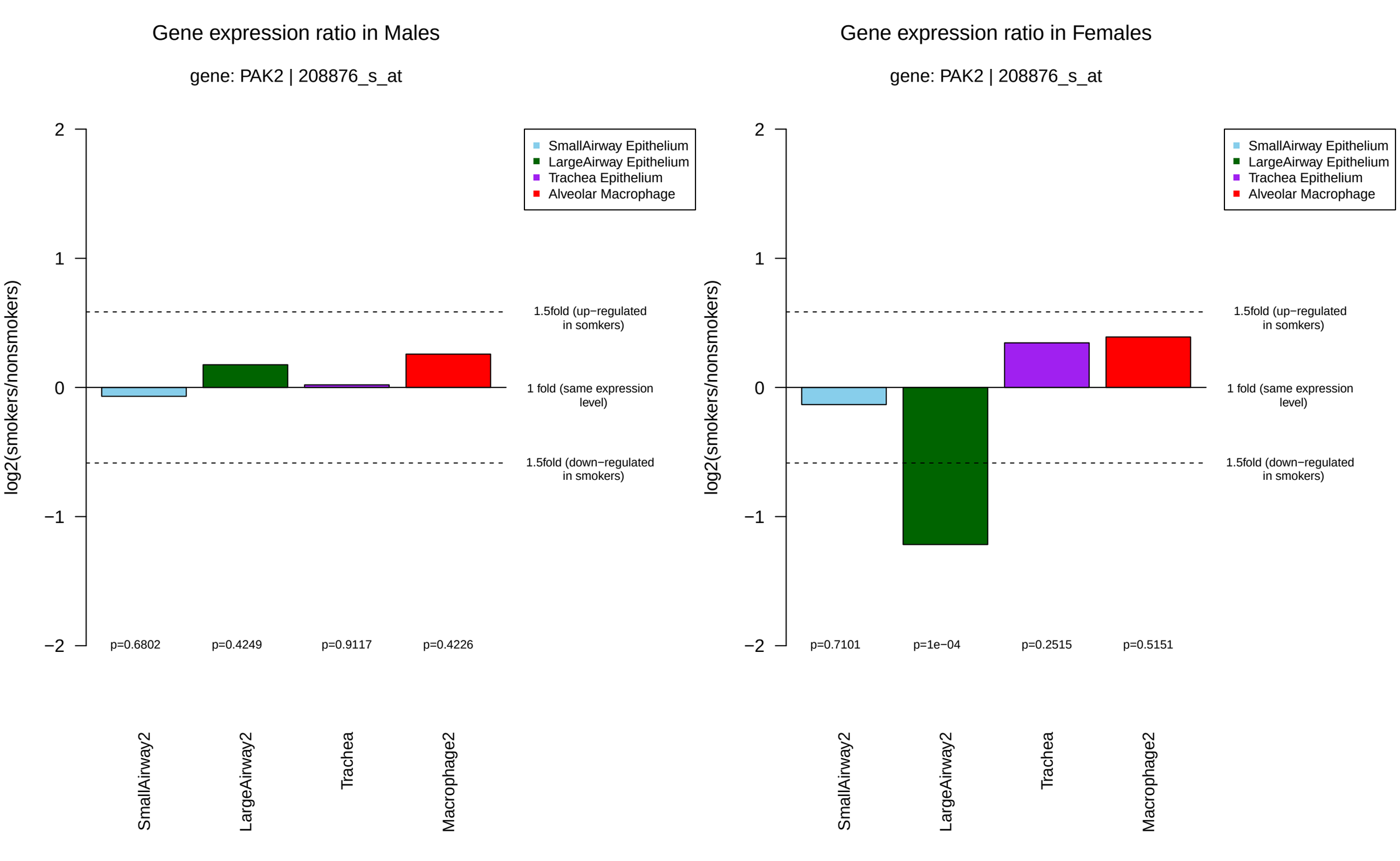
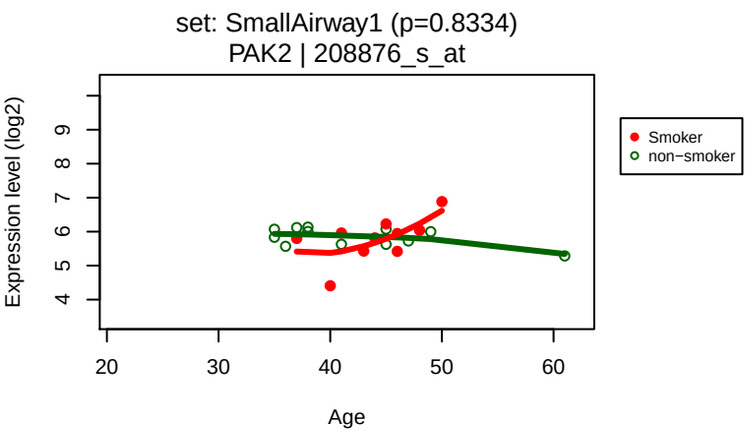
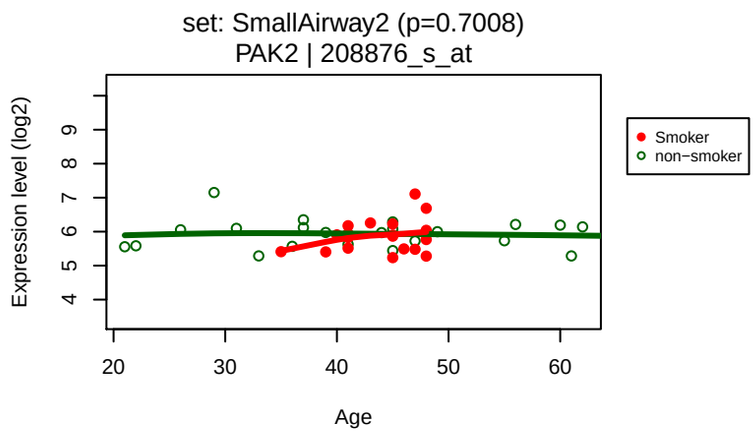
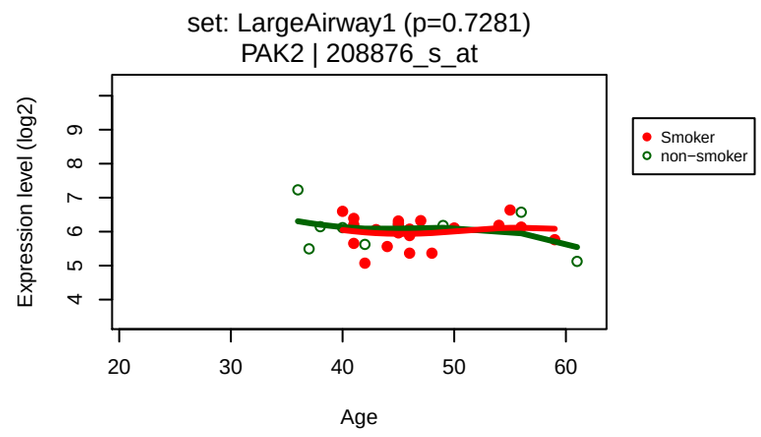

Gene expression of smokers and nonsmokers
gene: PAK2 | 208876_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8334
p=0.7008
p=0.1954
p=0.7281
p=0.0988
p=0.6609
p=0.5365
p=0.3119
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

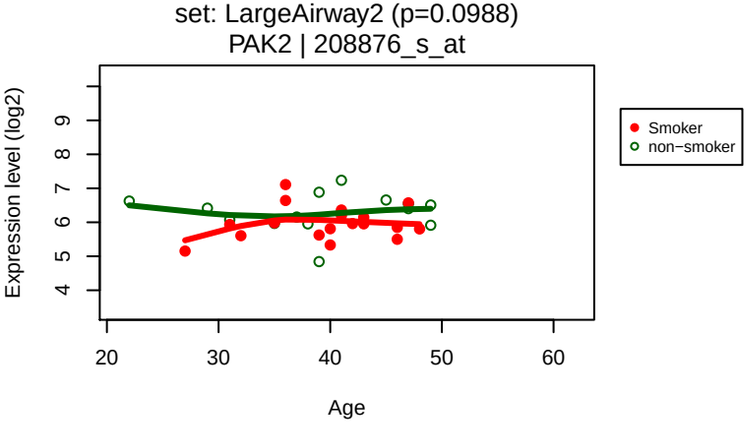

Gene expression ratio (smokers/nonsmokers)
gene: PAK2 | 208876_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8334
p=0.7008
p=0.1954
p=0.7281
p=0.0988
p=0.6609
p=0.5365
p=0.3119
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

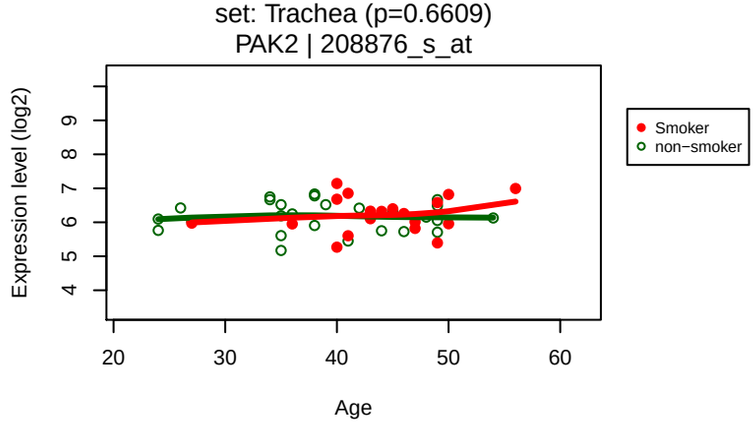

Gene expression ratio in Males
gene: PAK2 | 208876_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6802
p=0.4249
p=0.9117
p=0.4226
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK2 | 208876_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7101
p=1e−04
p=0.2515
p=0.5151
SmallAirway2
LargeAirway2
Trachea
Macrophage2

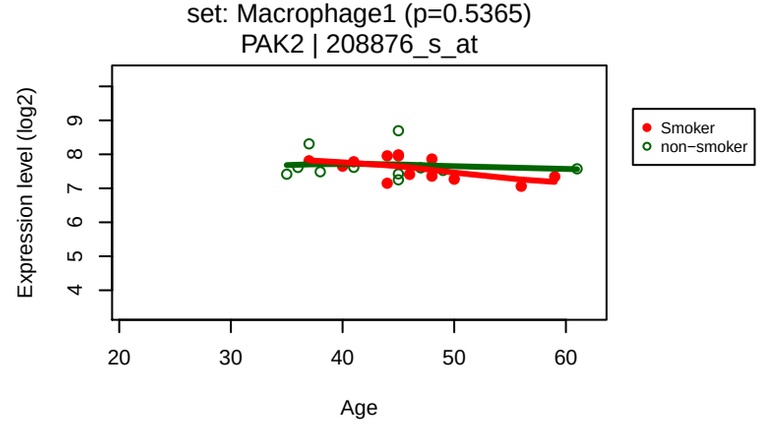

set: SmallAirway1 (p=0.8334)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker

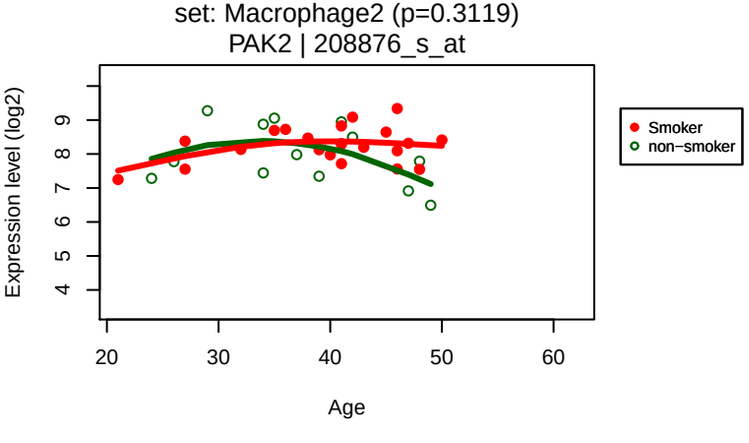

set: SmallAirway2 (p=0.7008)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker

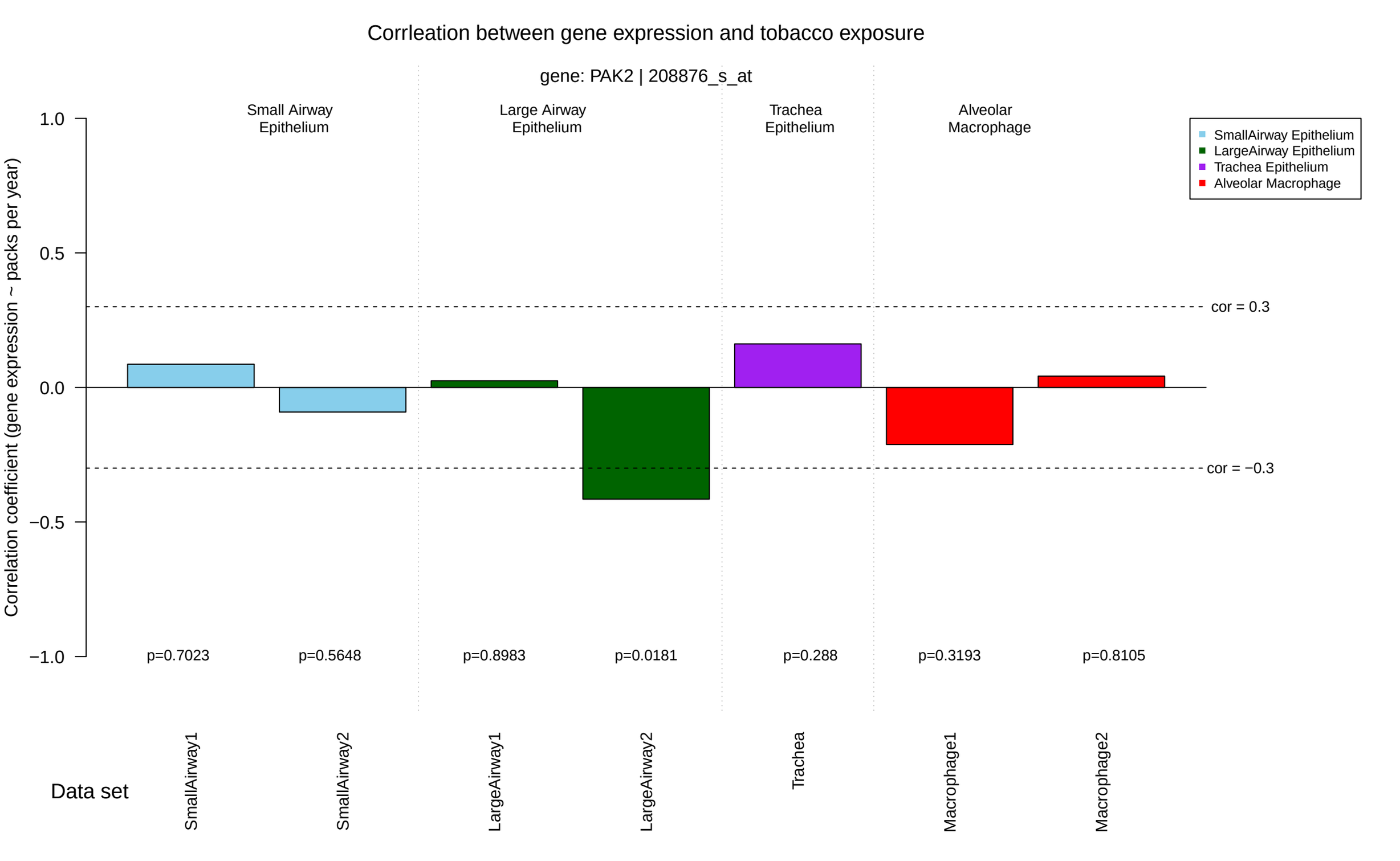

set: LargeAirway1 (p=0.7281)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0988)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6609)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5365)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3119)
PAK2 | 208876_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK2 | 208876_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7023
p=0.5648
p=0.8983
p=0.0181
p=0.288
p=0.3193
p=0.8105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2